# Assignment 5 solution

## 1. Part one

- **Interface**

```
interface ALSU_if(clk);
    input clk;
    logic cin, rst, red_op_A, red_op_B, bypass_A, bypass_B, direction, serial_in;
    logic [2:0] opcode;
    logic signed [2:0] A, B;
    logic [15:0] leds;
    logic signed [5:0] out;
endinterface
```

- **Top**

```
import uvm_pkg::*;
`include "uvm_macros.svh"

import alsu_test_pkg::*;    Could not f
module top();
    //? clk generation
    bit clk;
    always #1 clk = ~clk;
    //? DUT & interface instantiation
    ALSU_if alsuif (clk);
    ALSU DUT (
        .clk(alsuif.clk),
        .cin(alsuif.cin),
        .rst(alsuif.rst),
        .red_op_A(alsuif.red_op_A),
        .red_op_B(alsuif.red_op_B),
        .bypass_A(alsuif.bypass_A),
        .bypass_B(alsuif.bypass_B),
        .direction(alsuif.direction),
        .serial_in(alsuif.serial_in),
        .opcode(alsuif.opcode),
        .A(alsuif.A),
        .B(alsuif.B),
        .leds(alsuif.leds),
        .out(alsuif.out)
    );
    //? UVM run_test task
    initial begin
        run_test("alsu_test");
    end

endmodule
```

- **Alsu test class**

```
package alsu_test_pkg;
    import uvm_pkg::*;
    import alsu_env_pkg::*;    Could not find the package (alsu_env_pkg). Desi
    `include "uvm_macros.svh"

    class alsu_test extends uvm_test;
        //? factory register
        `uvm_component_utils(alsu_test)
        //? child object handle creation
        alsu_env env;    Invalid type 'alsu_env'. Please check the type of the
        //? constructor
        function new(string name="alsu_test", uvm_component parent = null);
            super.new(name, parent);
        endfunction
        //? phases
        //* build phase
        function void build_phase(uvm_phase phase);
            super.build_phase(phase);
            env = alsu_env::type_id::create("env",this);    Undefined variable
        endfunction
        //* run phase
        task run_phase(uvm_phase phase);
            super.run_phase(phase);
            phase.raise_objection(this);
            #10; `uvm_info("run_phase", "welcome to the UVM Env.", UVM_MEDIUM)
            phase.drop_objection(this);
        endtask

    endclass
endpackage
```

- **Alsu env class**

```
package alsu_env_pkg;
    import uvm_pkg::*;
    `include "uvm_macros.svh"

    class alsu_env extends uvm_env;
        //? factory register
        `uvm_component_utils(alsu_env)
        //? constructor
        function new(string name="alsu_env", uvm_component parent = null);
            super.new(name, parent);
        endfunction
    endclass
endpackage
```

- **Simulation result**

```
#
# UVM_INFO verilog_src/questa_uvm_pkg-1.2/src/questa_uvm_pkg.sv(277) @ 0: reporter [Questa UVM] QUESTA_UVM-1.2.3
# UVM_INFO verilog_src/questa_uvm_pkg-1.2/src/questa_uvm_pkg.sv(278) @ 0: reporter [Questa UVM]  questa_uvm::init(all)
# UVM_INFO @ 0: reporter [RNTST] Running test alsu_test...
# UVM_INFO alsu_test.sv(29) @ 100: uvm_test_top [run_phase] Inside the ALSU test
# UVM_INFO verilog_src/uvm-1.1d/src/base/uvm_objection.svh(1267) @ 100: reporter [TEST_DONE] 'run' phase is ready to proceed to the 'extract' phase
#
# --- UVM Report Summary ---
#
# ** Report counts by severity
# UVM_INFO :    5
# UVM_WARNING :    0
# UVM_ERROR :    0
# UVM_FATAL :    0
# ** Report counts by id
# [Questa UVM]     2
# [RNTST]     1
# [TEST_DONE]     1
# [run_phase]     1
# ** Note: $finish    : C:/questasim64_2021.1/win64/../verilog_src/uvm-1.1d/src/base/uvm_root.svh(430)
```

## 2. Part two

- **Alsu test class**

```
package alsu_test_pkg;
    import uvm_pkg::*;
    import alsu_env_pkg::*;     Could not find the package (alsu_env_pkg). Design reac
    `include "uvm_macros.svh"

    class alsu_test extends uvm_test;
        //? factory register
        `uvm_component_utils(alsu_test)
        //? child object handle creation
        alsu_env env;     Invalid type 'alsu_env'. Please check the type of the variab
        // virtual interface
        virtual ALSU_if alsu_test_vif;
        //? constructor
        function new(string name="alsu_test", uvm_component parent = null);
            super.new(name, parent);
        endfunction
        //? phases
        //* build phase
        function void build_phase(uvm_phase phase);
            super.build_phase(phase);
            env = alsu_env::type_id::create("env",this);     Undefined variable: 'env'
            uvm_config_db #(virtual ALSU_if)::get(this,"","ALSU_IF", alsu_test_vif);
            uvm_config_db #(virtual ALSU_if)::set(this,"*","INTF",alsu_test_vif);
        endfunction
        //* run phase
        task run_phase(uvm_phase phase);
            super.run_phase(phase);
            phase.raise_objection(this);
            #100; `uvm_info("run_phase", "Inside the ALSU test", UVM_MEDIUM)
            phase.drop_objection(this);
        endtask

    endclass
endpackage
```

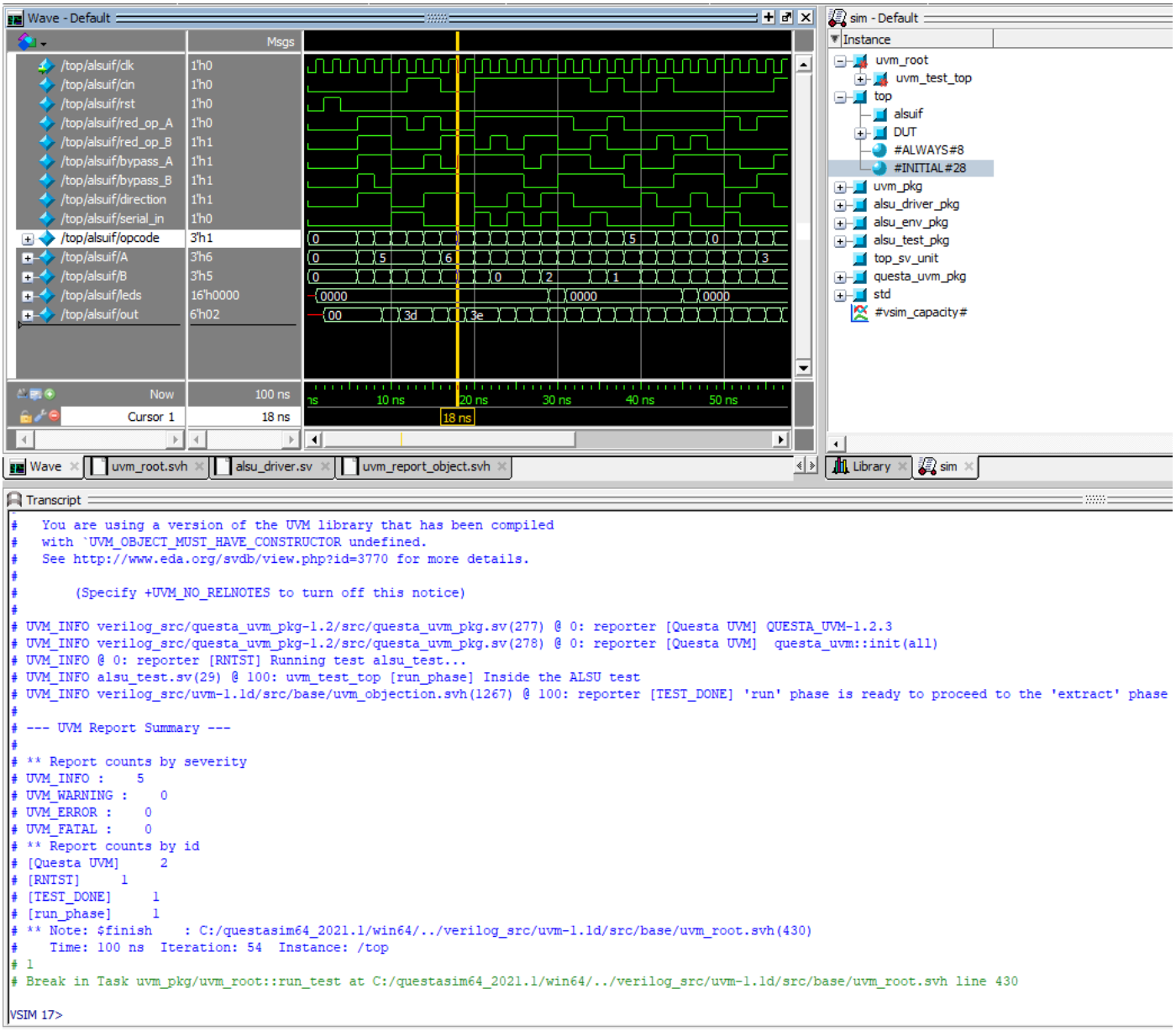
- **Alsu env class**

```
package alsu_env_pkg;
    import uvm_pkg::*;
    import alsu_driver_pkg::*;    Could not find the package (alsu_driver_p
    `include "uvm_macros.svh"

    class alsu_env extends uvm_env;
        //? factory register
        `uvm_component_utils(alsu_env)
        //? child object handle creation
        alsu_driver driver;    Invalid type 'alsu_driver'. Please check the
        //? constructor
        function new(string name="alsu_env", uvm_component parent = null);
            super.new(name, parent);
        endfunction
        //?phases
        //* build phase
        function void build_phase(uvm_phase phase);
            super.build_phase(phase);
            driver = alsu_driver::type_id::create("driver",this);    Undefi
    endfunction
    endclass
endpackage
```

- **Alsu driver class**

```
package alsu_driver_pkg;
  import uvm_pkg::*;
  `include "uvm_macros.svh"
  class alsu_driver extends uvm_driver;
    `uvm_component_utils(alsu_driver)

    virtual ALSU_if alsu_driver_vif;

    function new(string name = "alsu_driver", uvm_component parent = null);
      super.new(name, parent);
    endfunction

    function void build_phase(uvm_phase phase);
      super.build_phase(phase);
      uvm_config_db #(virtual ALSU_if)::get(this,"","INTF", alsu_driver_vif);
    endfunction

    task run_phase(uvm_phase phase);
        super.run_phase(phase);
        alsu_driver_vif.rst = 0;
        alsu_driver_vif.cin = 0;
        alsu_driver_vif.red_op_A = 0;
        alsu_driver_vif.red_op_B = 0;
        alsu_driver_vif.opcode = 0;
        alsu_driver_vif.bypass_A = 0;
        alsu_driver_vif.bypass_B = 0;
        alsu_driver_vif.direction = 0;
        alsu_driver_vif.serial_in = 0;
        alsu_driver_vif.A = 0;
        alsu_driver_vif.B = 0;
        @(negedge alsu_driver_vif.clk) alsu_driver_vif.rst = 1;
        @(negedge alsu_driver_vif.clk) alsu_driver_vif.rst = 0;
        forever begin
            @(negedge alsu_driver_vif.clk);
            alsu_driver_vif.cin = $random;
            alsu_driver_vif.red_op_A = $random;
            alsu_driver_vif.red_op_B = $random;
            alsu_driver_vif.opcode = $random;
            alsu_driver_vif.bypass_A = $random;
            alsu_driver_vif.bypass_B = $random;
            alsu_driver_vif.direction = $random;
            alsu_driver_vif.serial_in = $random;
            alsu_driver_vif.A = $random;
            alsu_driver_vif.B = $random;
        end
    endtask
  endclass
endpackage
```

- **Simulation results**




## 3. Part three

- **Configuration object class**

```
package alsu_config_obj_pkg;
    import uvm_pkg::*;
    `include "uvm_macros.svh"

    class alsu_config_obj extends uvm_object;
        `uvm_object_utils(alsu_config_obj)

        virtual ALSU_if alsu_config_vif;

        function new(string name = "alsu_config_obj");
        super.new(name);
        endfunction

    endclass
endpackage
```

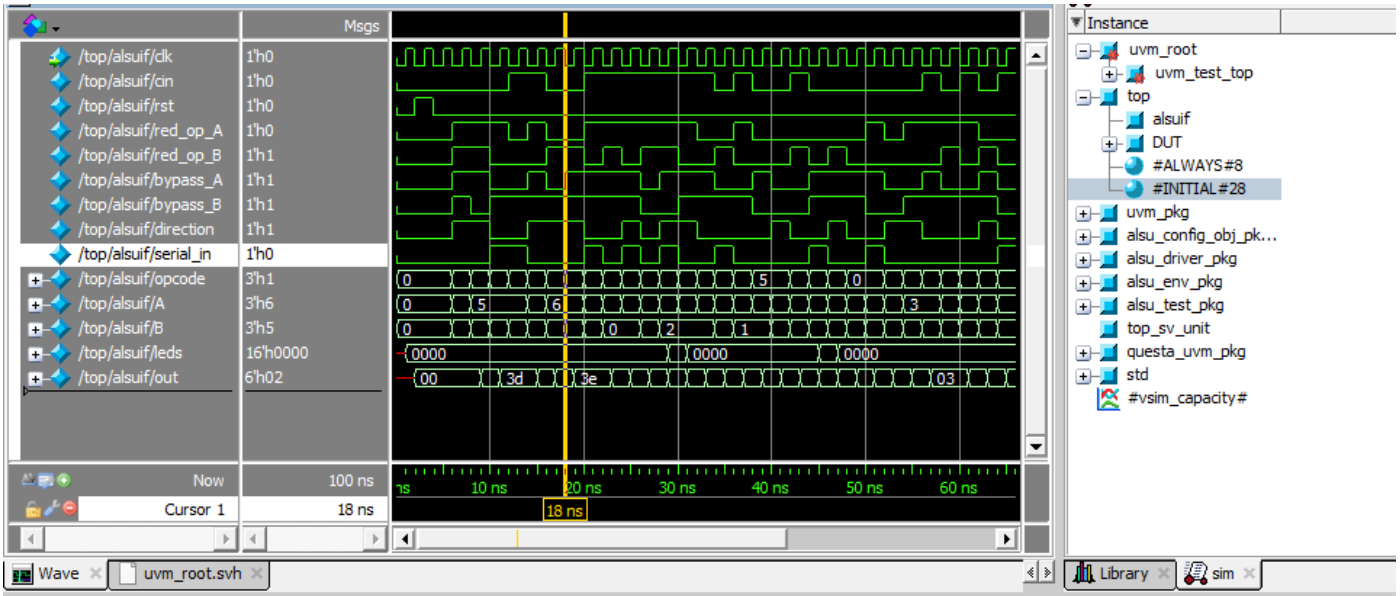
- **Alsu test class**

```
package alsu_test_pkg;
    import uvm_pkg::*;
    import alsu_env_pkg::*;    Could not find the package (alsu_env_pkg). Design read will continue, but exp
    import alsu_config_obj_pkg::*;    Could not find the package (shift_reg_config_pkg). Design read will co
    `include "uvm_macros.svh"

    class alsu_test extends uvm_test;
        //? factory register
        `uvm_component_utils(alsu_test)
        //? child object handle creation
        alsu_env env;    Invalid type 'alsu_env'. Please check the type of the variable 'env'.
        //? configuration object handle creation
        alsu_config_obj alsu_config_obj_test;    Invalid type 'alsu_config_obj'. Please check the type of th
        //? constructor
        function new(string name="alsu_test", uvm_component parent = null);
            super.new(name, parent);
        endfunction
        //? phases
        //* build phase
        function void build_phase(uvm_phase phase);
            super.build_phase(phase);
            env = alsu_env::type_id::create("env",this);    Undefined variable: 'env'.
            alsu_config_obj_test = alsu_config_obj::type_id::create("alsu_config_obj_test");    Undefined va
            uvm_config_db #(virtual ALSU_if)::get(this,"","ALSU_IF", alsu_config_obj_test.alsu_config_vif);
            uvm_config_db #(alsu_config_obj)::set(this,"*","CFG",alsu_config_obj_test);
        endfunction
        //* run phase
        task run_phase(uvm_phase phase);
            super.run_phase(phase);
            phase.raise_objection(this);
            #100; `uvm_info("run_phase", "Inside the ALSU test", UVM_MEDIUM)
            phase.drop_objection(this);
        endtask

    endclass
endpackage
```

- **Alsu driver class**

```
package alsu_driver_pkg;
  import uvm_pkg::*;
  import alsu_config_obj_pkg::*;    Could not find the package (alsu_config_obj_pkg). Desig
  `include "uvm_macros.svh"
  class alsu_driver extends uvm_driver;
    `uvm_component_utils(alsu_driver)

    alsu_config_obj alsu_config_obj_driver;    Invalid type 'alsu_config_obj'. Please check
    virtual ALSU_if alsu_driver_vif;

    function new(string name = "alsu_driver", uvm_component parent = null);
      super.new(name, parent);
    endfunction

    function void build_phase(uvm_phase phase);
      super.build_phase(phase);
      alsu_config_obj_driver = alsu_config_obj::type_id::create("alsu_config_obj_driver");
      uvm_config_db #(alsu_config_obj)::get(this,"","CFG", alsu_config_obj_driver);
    endfunction

    function void connect_phase(uvm_phase phase);
      super.build_phase(phase);
      alsu_driver_vif = alsu_config_obj_driver.alsu_config_vif;
    endfunction

    task run_phase(uvm_phase phase);
        super.run_phase(phase);
        alsu_driver_vif.rst = 0;
        alsu_driver_vif.cin = 0;
        alsu_driver_vif.red_op_A = 0;
        alsu_driver_vif.red_op_B = 0;
        alsu_driver_vif.opcode = 0;
        alsu_driver_vif.bypass_A = 0;
        alsu_driver_vif.bypass_B = 0;
        alsu_driver_vif.direction = 0;
        alsu_driver_vif.serial_in = 0;
        alsu_driver_vif.A = 0;
        alsu_driver_vif.B = 0;
        @(negedge alsu_driver_vif.clk) alsu_driver_vif.rst = 1;
        @(negedge alsu_driver_vif.clk) alsu_driver_vif.rst = 0;
        forever begin
            @(negedge alsu_driver_vif.clk);
            alsu_driver_vif.cin = $random;
            alsu_driver_vif.red_op_A = $random;
            alsu_driver_vif.red_op_B = $random;
            alsu_driver_vif.opcode = $random;
            alsu_driver_vif.bypass_A = $random;
            alsu_driver_vif.bypass_B = $random;
            alsu_driver_vif.direction = $random;
            alsu_driver_vif.serial_in = $random;
            alsu_driver_vif.A = $random;
            alsu_driver_vif.B = $random;
        end
    endtask
  endclass
endpackage
```

- **Simulation results**




Transcript

```
#   See http://www.eda.org/svdb/view.php?id=3770 for more details.
#
#       (Specify +UVM_NO_RELNOTES to turn off this notice)
#
# UVM_INFO verilog_src/questa_uvm_pkg-1.2/src/questa_uvm_pkg.sv(277) @ 0: reporter [Questa UVM] QUESTA_UVM-1.2.3
# UVM_INFO verilog_src/questa_uvm_pkg-1.2/src/questa_uvm_pkg.sv(278) @ 0: reporter [Questa UVM]  questa_uvm::init(all)
# UVM_INFO @ 0: reporter [RNTST] Running test alsu_test...
# UVM_WARNING @ 0: uvm_test_top.env.driver [UVM_DEPRECATED] build()/build_phase() has been called explicitly, outside of the phasing sys
# UVM_INFO alsu_test.sv(31) @ 100: uvm_test_top [run_phase] Inside the ALSU test
# UVM_INFO verilog_src/uvm-1.1d/src/base/uvm_objection.svh(1267) @ 100: reporter [TEST_DONE] 'run' phase is ready to proceed to the 'ext
#
# --- UVM Report Summary ---
#
# ** Report counts by severity
# UVM_INFO :    5
# UVM_WARNING :    1
# UVM_ERROR :    0
# UVM_FATAL :    0
# ** Report counts by id
# [Questa UVM]     2
# [RNTST]     1
# [TEST_DONE]     1
# [UVM_DEPRECATED]     1
# [run_phase]     1
# ** Note: $finish    : C:/questasim64_2021.1/win64/../verilog_src/uvm-1.1d/src/base/uvm_root.svh(430)
#    Time: 100 ns  Iteration: 54  Instance: /top
# 1
# Break in Task uvm_pkg/uvm_root::run_test at C:/questasim64_2021.1/win64/../verilog_src/uvm-1.1d/src/base/uvm_root.svh line 430

VSIM 2>
```